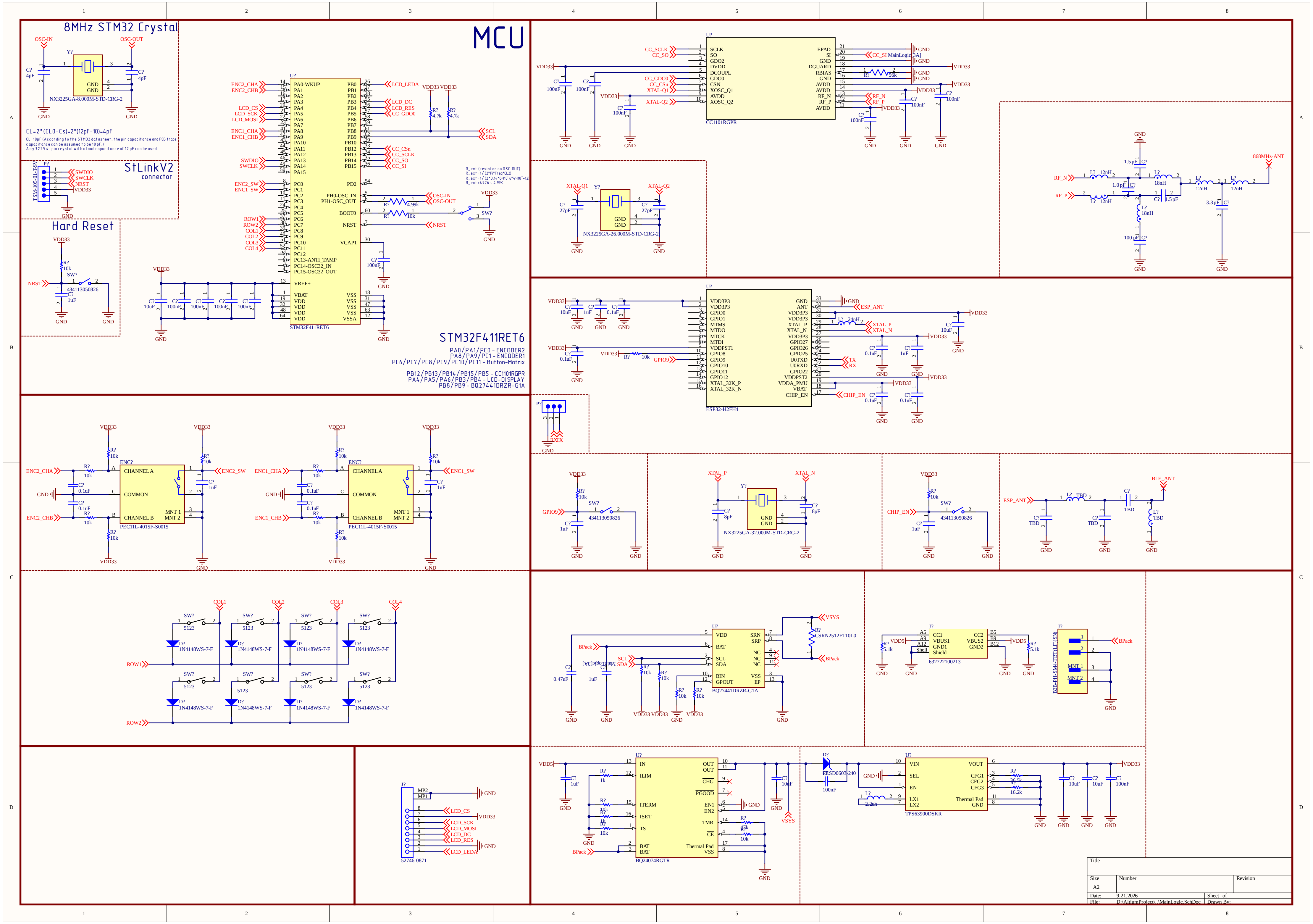
MCU
8MHz STM32 Crystal
OSC-IN
OSC-OUT
Y?
NX3225GA-8.000M-STD-CRG-2
C?
4pF
GND
CL=2*(CL0-Cs)=2*(12pF-10)=4pF
CL=10pF (According to the STM32 datasheet, the pin capacitance and PCB trace capacitance can be assumed to be 10 pF.)
Any 3225 4-pin crystal with a load capacitance of 12 pF can be used.
StLinkV2
connector
P?
TSM-105-01-T-SV
SWDIO
SWCLK
NRST
VDD33
Hard Reset
R?
10k
SW?
434113050826
1uF
U?
STM32F411RET6
PA0-WKUP
PB0
PB1
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13-ANTI_TAMP
PC14-OSC32_IN
PC15-OSC32_OUT
VREF+
VBAT
VDD
VSS
VSSA
PD2
PH0-OSC_IN
PH1-OSC_OUT
BOOT0
NRST
VCAP1
ENC2_CHA
ENC2_CHB
LCD_CS
LCD_SCK
LCD_MOSI
ENC1_CHA
ENC1_CHB
ENC2_SW
ENC1_SW
ROW1
ROW2
COL1
COL2
COL3
COL4
LCD_LEDA
LCD_DC
LCD_RES
CC_GDO0
CC_CSn
CC_SCLK
CC_SO
CC_SI
SCL
SDA
4.7k
4.99k
10k
100nF
10uF
R_ext (resistor on OSC-OUT)
R_ext=1/(2*Pi*Freq*CL2)
R_ext=1/(2*3.14*8410^6*4410^-12)
R_ext=4976 ~ 4.99K
STM32F411RET6
PA0/PA1/PC0 - ENCODER2
PA8/PA9/PC1 - ENCODER1
PC6/PC7/PC8/PC9/PC10/PC11 - Button-Matrix
PB12/PB13/PB14/PB15/PB5 - CC1101RGPR
PA4/PA5/PA6/PB3/PB4 - LCD-DISPLAY
PB8/PB9 - BQ27441DRZR-G1A
CC1101RGPR
SCLK
SO
GDO2
DVDD
DCOUPL
GDO0
CSN
XOSC_Q1
AVDD
XOSC_Q2
EPAD
SI
GND
DGUARD
RBIAS
RF_N
RF_P
CC_SI MainLogic[3A]
XTAL-Q1
XTAL-Q2
56k
NX3225GA-26.000M-STD-CRG-2
27pF
868MHz-ANT
1.5 pF
12nH
18nH
1.0 pF
3.3 pF
100 pF
ESP32-H2FH4
VDD3P3
GPIO0
GPIO1
MTMS
MTDO
MTCK
MTDI
VDDPST1
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
XTAL_32K_P
XTAL_32K_N
ANT
XTAL_P
XTAL_N
GPIO27
GPIO26
GPIO25
U0TXD
U0RXD
GPIO22
VDDPST2
VDDA_PMU
VBAT
CHIP_EN
ESP_ANT
TX
RX
24nH
0.1uF
RXTX
ENC?
CHANNEL A
COMMON
CHANNEL B
MNT 1
MNT 2
PEC11L-4015F-S0015
NX3225GA-32.000M-STD-CRG-2
8pF
BLE_ANT
TBD
5123
1N4148WS-7-F
U?
BQ27441DRZR-G1A
VDD
BAT
SCL
SDA
BIN
GPOUT
SRN
SRP
NC
VSS
EP
CSRN2512FT10L0
VSYS
BPack
0.47uF
J?
632722100213
CC1
VBUS1
GND1
Shell
CC2
VBUS2
GND2
VDD5
5.1k
B2B-PH-SM4-TBT(LF)(SN)
MNT 1
MNT 2
BQ24074RGTR
IN
ILIM
ITERM
ISET
TS
OUT
CHG
PGOOD
EN1
EN2
TMR
CE
Thermal Pad
PESD0603-240
TPS63900DSKR
VIN
SEL
EN
LX1
LX2
VOUT
CFG1
CFG2
CFG3
Thermal Pad
2.2uh
36.5k
16.2k
52746-0871
MP2
MP1
Title
Size
A2
Number
Revision
Date: 9.21.2026
Sheet of
File: D:\AltiumProject\..\MainLogic.SchDoc
Drawn By:

Comment	Description	Designator	Footprint	LibRef	Quantity
0.1uF		C?	CAPC2013X94X50NL20T25	CMP-2007-04406-1	4
0.1uF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 0.10uF, X7S, 22%, 10%, 50V	C?	FP-GRM155-0_05-MFG	CMP-06035-019038-1	5
0.47uF		C?	CAPC2013X94X50NL20T25	CMP-2007-04406-1	1
1.0 pF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 470pF, C0G, 30ppm/°C, 5%, 50V	C?	FP-GRM155-0_05-IPC_A	CMP-2008-04370-2	1
1uF		C?	CAPC2013X94X50NL20T25	CMP-2007-04406-1	2
1uF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 0.10uF, X7S, 22%, 10%, 50V	C?	FP-GRM155-0_05-MFG	CMP-06035-019038-1	7
1.5 pF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 470pF, C0G, 30ppm/°C, 5%, 50V	C?	FP-GRM155-0_05-IPC_A	CMP-2008-04370-2	2
3.3 pF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 470pF, C0G, 30ppm/°C, 5%, 50V	C?	FP-GRM155-0_05-IPC_A	CMP-2008-04370-2	1
4pF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 470pF, C0G, 30ppm/°C, 5%, 50V	C?	FP-GRM155-0_05-IPC_A	CMP-2008-04370-2	2
8pF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 470pF, C0G, 30ppm/°C, 5%, 50V	C?	FP-GRM155-0_05-IPC_A	CMP-2008-04370-2	2
10uF		C?	CAPC2013X94X50NL20T25	CMP-2007-04406-1	3
10uF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 0.10uF, X7S, 22%, 10%, 50V	C?	FP-GRM155-0_05-MFG	CMP-06035-019038-1	3
27pF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 470pF, C0G, 30ppm/°C, 5%, 50V	C?	FP-GRM155-0_05-IPC_A	CMP-2008-04370-2	2
100 pF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 470pF, C0G, 30ppm/°C, 5%, 50V	C?	FP-GRM155-0_05-IPC_A	CMP-2008-04370-2	1
100nF		C?	CAPC2013X94X50NL20T25	CMP-2007-04406-1	2
100nF	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 0.10uF, X7S, 22%, 10%, 50V	C?	FP-GRM155-0_05-MFG	CMP-06035-019038-1	11
TBD	Chip Multilayer Ceramic Capacitors for General Purpose, 0402, 0.10uF, X7S, 22%, 10%, 50V	C?	FP-GRM155-0_05-MFG	CMP-06035-019038-1	3
1N4148WS-7-F	Fast Switching Diode, 75 V, 0.3 A, -65 to 150 degC, 2-Pin SOD323, RoHS, Tape and Reel	D?	DIOD-SOD323-2-A_V	CMP-1667-00004-1	8
PESD0603-240	ESD Protector Overvoltage Protection Device, 0.25 pF, 24 V, -55 to 125 degC, 2-Pin SMD (0603), RoHS, Tape and Reel	D?	TECO-0603_V	CMP-2000-06889-1	1
PEC11L-4015F-S0015	ROTARY ENCODER MECHANICAL 15PPR	ENC?	FP-PEC11L-4015F-S0015-MFG	CMP-07421-000033-1	2
52746-0871	0.5 mm Pitch Easy-On(TM) Type FPC Connector, 2.0 mm Mated Height, Right Angle, ZIF, SMT, Bottom Contact, 8 Circuits, -20 to 85 degC, ELV and RoHS Compliant, Tape and Reel	J?	MOLX-52746-0871	CMP-1241-00370-1	1
632722100213	Usb Conn, Type-c, Rcpt, 6POS Rohs Compliant: Yes	J?	632722100213	CMP-03913-147573503-1	1
B2B-PH-SM4-TBT(LF)(SN)	CONN HEADER SMD 2POS 2MM	J?	FP-B2B-PH-SM4-TBT_LF_SN-MFG	CMP-17439-000005-1	1
2.2uh	Automotive Inductors, High Temperature (180 °C) Series 2.2uH 25.5A 3.67mΩ 20%	L?	FP-IHLP-5050FD-FP_3_226x4-MFG	CMP-02424-000098-1	1
12nH	Multilayer type Inductor for Automotive 12nH ±3% 0.24Ω 450mA 0402	L?	FP-LQG15WH_02-IPC_C	CMP-06042-014976-1	4
18nH	Multilayer type Inductor for Automotive 12nH ±3% 0.24Ω 450mA 0402	L?	FP-LQG15WH_02-IPC_C	CMP-06042-014976-1	2
24nH	Multilayer type Inductor for Automotive 24nH ±3% 1.2Ω 230mA 0402	L?	FP-LQG15WH_02-MFG	CMP-06042-002765-1	1
TBD	Multilayer type Inductor for Automotive 24nH ±3% 1.2Ω 230mA 0402	L?	FP-LQG15WH_02-MFG	CMP-06042-002765-1	2
MTSW-103-08-G-S-229-RA	Modified .025" SQ Post Header, Through-hole, Right Angle, -55 to 125 degC, 2.54 mm Pitch, 3-Pin, Male, RoHS	P?	SMTC-MTSW-103-08-X-S-229-RA	CMP-1024-00100-1	1
TSM-105-01-T-SV	0.025" SQ Post Header, Surface Mount, Vertical, -55 to 105 degC, 2.54 mm Pitch, 5-Pin, Male, RoHS	P?	SMTC-TSM-105-01-X-SV	CMP-1024-00239-1	1
1k	1K18 0.125W 1% 0805 (2012 Metric) SMD	R?	RESC0805(2012)_M	CMP-1013-00517-1	2
4.7k	1K18 0.125W 1% 0805 (2012 Metric) SMD	R?	RESC0805(2012)_M	CMP-1013-00517-1	2
4.99k	CRGCQ 0402 56K 1%	R?	FP-CRGCQ0402-IPC_C	CMP-03211-009184-1	1
5.1k	1K18 0.125W 1% 0805 (2012 Metric) SMD	R?	RESC0805(2012)_M	CMP-1013-00517-1	2
10k	CRGCQ 0402 56K 1%	R?	FP-CRGCQ0402-IPC_C	CMP-03211-009184-1	1
10k	1K18 0.125W 1% 0805 (2012 Metric) SMD	R?	RESC0805(2012)_M	CMP-1013-00517-1	21
16.2k	1K18 0.125W 1% 0805 (2012 Metric) SMD	R?	RESC0805(2012)_M	CMP-1013-00517-1	1
36.5k	1K18 0.125W 1% 0805 (2012 Metric) SMD	R?	RESC0805(2012)_M	CMP-1013-00517-1	1
47k	1K18 0.125W 1% 0805 (2012 Metric) SMD	R?	RESC0805(2012)_M	CMP-1013-00517-1	1
56k	CRGCQ 0402 56K 1%	R?	FP-CRGCQ0402-IPC_C	CMP-03211-009184-1	1
CSRN2512FT10L0	RES 0.01 OHM 1% 2W 2512	R?	FP-CSRN2512-IPC_B	CMP-26527-000109-1	1
	SWITCH SLIDE SPDT 300MA 6V JS102011SAQN	SW?	FP-JS102011SAQN-MFG	CMP-2000-06501-2	1
5123	Description	SW?	Kailh_socket_MX	5123	8
434113050826	Switch Tactile Spst-no 0.05A 12V	SW?	434113050826	CMP-1460-00003-6	3
BQ24074RGTR	IC LI-ION CHARGE MGMT 16-QFN	U?	FP-RGT0016C-IPC_B	CMP-0321-00220-3	1
BQ27441DRZR-G1A	IC BATT FUEL GAUGE LI-ION 12SON	U?	FP-DRZ0012A-MFG	CMP-04911-000008-1	1
CC1101RGPR	Low-power Sub-1 GHz wireless transceiver 20-QFN -40 to 85	U?	RGP20_2P4X2P4	CC1101RGPR	1
ESP32-H2FH4	Description	U?	QFN324X4_EXP	ESP32-H2FH4	1
STM32F411RET6	ARM Cortex-M4 32-bit MCU+FPU, 512 KB Flash, 128 KB RAM, 50 I/Os, 64-Pin LQFP, -40 to 85 degC, Tray	U?	STM-LQFP64_L	CMP-0243-00169-1	1
TPS63900DSKR	1.8-V TO 5.5-V, 75-NA IQ BUCK-BO	U?	FP-DSK0010A-IPC_C	CMP-04918-000700-1	1
NX3225GA-8.000M-STD-CRG-2	CRYSTAL 16.0000MHZ 8PF SMD	Y?	FP-NX3225GA-MFG	CMP-200220-000007-1	1
NX3225GA-26.000M-STD-CRG-2	CRYSTAL 16.0000MHZ 8PF SMD	Y?	FP-NX3225GA-MFG	CMP-200220-000007-1	1
NX3225GA-32.000M-STD-CRG-2	CRYSTAL 16.0000MHZ 8PF SMD	Y?	FP-NX3225GA-MFG	CMP-200220-000007-1	1